

SpatialVerse

PIONEER IN AI DATA SOLUTIONS

Elevate Intelligence of AI Agents



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INTRODUCTION

About SpatialVerse

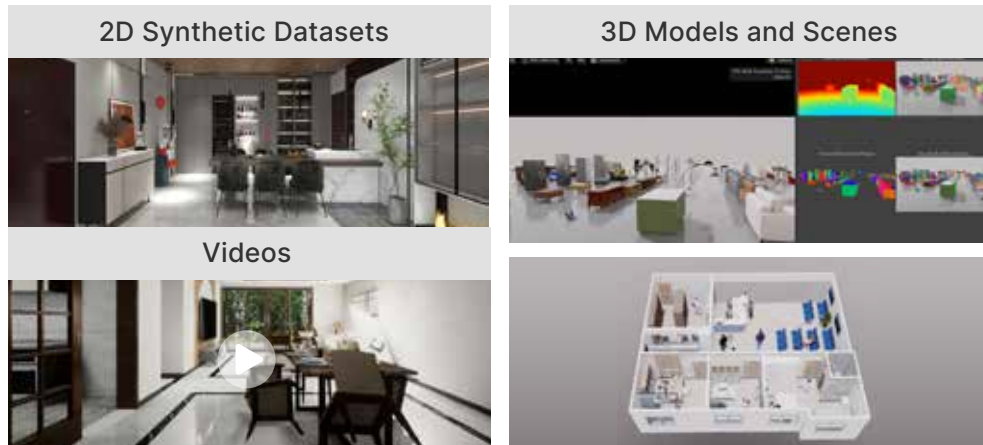
SpatialVerse, a professional data service provider incubated by Manycore Tech & Technology's innovation laboratory. Drawing upon the richness of Kujiale's superb Interior Design Solution globally, we offer an array of dataset services, including Interior Synthetic Data Generation and 3D Virtual Assets Creation vitally, facilitating expedited progress in diverse industries' development endeavors.

Mission

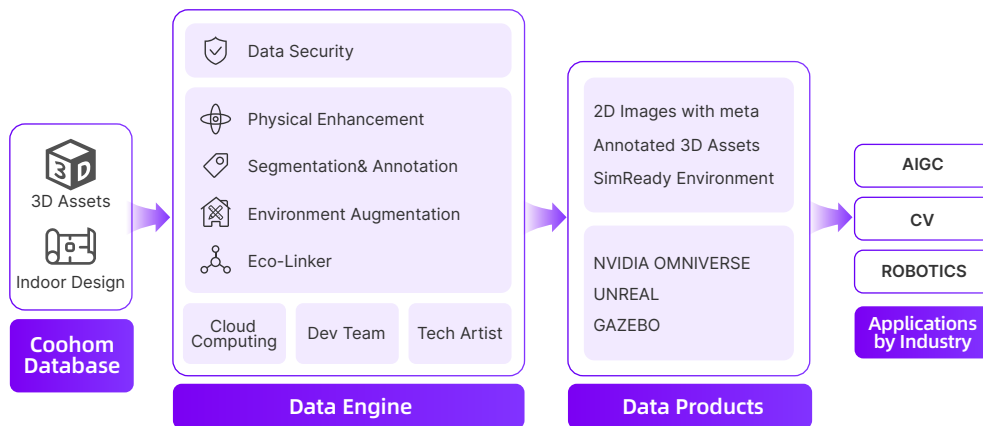
Utilizing the characteristic of diminishing marginal costs through synthetic datasets, and leveraging a systematic service framework that is toolized and customized, we provide Interior 2D/3D datasets with realistic physical properties to accelerate the advancement of the AI industry.

TECHNOLOGY

Our Products



Our Core Capabilities



Model Library

- 270 million 3D models, including furnitures, appliances, animals, household items, etc.
- 18 million interior design schemes, including houses, supermarkets, restaurants, offices, etc.



Physical Enhancement

- Physical parameters: density, friction, elasticity, damping, etc.
- Physical constraints: hinge, slide rail, roller, etc.



Segmentation & Annotation

- Combined automatic and manual annotation
- Enhanced efficiency and cost-effectiveness
- Labels: semantics, materials, and spatial states, etc.



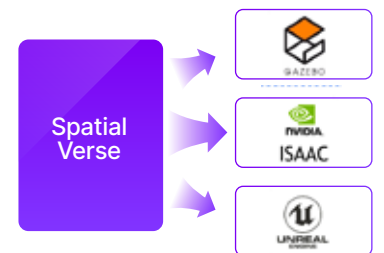
Environment Augmentation

- Floor plan design
- Lighting design
- Material variations



Eco-Linker

- Multi-channel data output
- Reduction of data leakage risk
- Avoidance of contact with real user data



Data Security Assurance

- End-to-end data security management
- Avoidance of contact with real user data
- Reduction of data leakage risk



Our Data Processing Tools

Scene Design Tool

The scene design tool is the most basic solution design platform of Coohom, It can meet various design needs, including house type, soft decoration, hard decoration, lighting, etc., and help to create diverse and one-stop design solutions.



Data Annotation Tool

An efficient data management platform that enables labeling of various information such as the overall model, components, materials, location, and status.



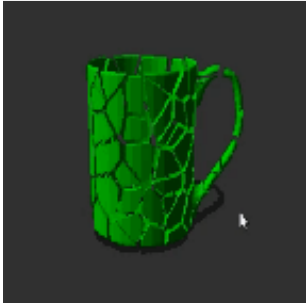
Lighting Simulation Tool

It achieves a rich lighting effect, allowing the scene to achieve various situational changes, such as the time change effect of morning and sunset, as well as the change control of individual light sources.



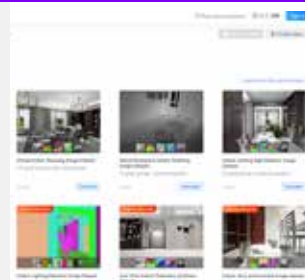
Model Enhancement Tool

We can achieve the flexible change of the model, such as the morphological change of the fabric model, the ground scattering of the liquid model, and the fragmentation of the glass model. Through the strengthening at the physical level, a simple model also has diversity.



Synthetic Image Generation Tool

KOOLAI is an AI training synthetic data platform for indoor intelligent agent cognition, relying on Kujiale's largest global 3D scene data resources in the world and backed by the Koolab laboratory, to provide diverse synthetic data of indoor scenes, making the intelligent agent more intelligent.



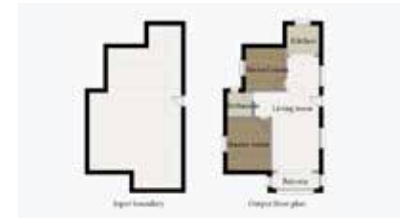
RESEARCH

Academic Achievements



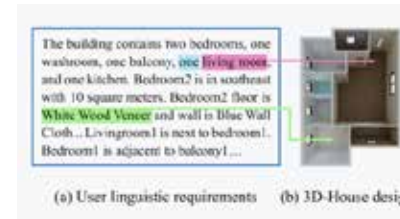
BMVC 2018

InteriorNet: Mega-scale Multi-sensor Photo-realistic Indoor Scenes Dataset



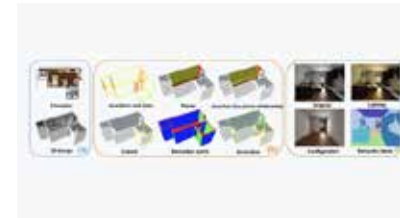
SIGGRAPH Asia 2019

Data-driven Interior Plan Generation for Residential Buildings



CVPR 2020

Intelligent Home 3D: Automatic 3D-House Design from Linguistic Descriptions Only



CCV 2020

Structured3D: A Large Photo-realistic Dataset for Structured 3D Modeling



CVPR 2021

Learning to Recommend Frame for Interactive Video Object Segmentation in the Wild



CGF 2022

MINERVAS: Massive Interior Environments Virtual Synthesis

Academic Collaboration

SPEAR:

A Simulator for Photorealistic Embodied AI Research



Interactive simulators are increasingly vital for training embodied artificial intelligence (AI) systems. However, current simulators suffer from restricted content diversity, physical interactivity, and visual fidelity. In response to these limitations, Intel Labs has partnered with the Computer Vision Center in Spain, Kujiale in China, and the Technical University of Munich to create the Simulator for Photorealistic Embodied AI Research (SPEAR). This advanced simulation platform significantly accelerates the training and validation of embodied agents across an expanding array of tasks and domains, directly benefiting the embodied AI developer community.

With its wide array of photorealistic indoor environments, SPEAR caters to diverse household navigation and manipulation tasks. It aims to advance research and commercial applications in household robotics and manufacturing, including human-robot interaction scenarios and digital twin applications.



Scan QR code to view the full article.

OUR SOLUTIONS

SpatialVerse utilizes Synthetic Data technology, its self-developed graphic rendering engine, and cloud computing platform to optimize indoor environments and model elements at scale. These customized dataset outputs address diverse industry needs.



High Quality



Cost Effectiveness



Privacy Assurance



Expanded Database



Automatic Labeling Tool

AI Agent Understanding

Relying on the rich data generation experience in the field of smart home, SpatialVerse helps indoor environmental smart devices, such as sweeping robots, care robots, and humanoid robots, to achieve technological breakthroughs in object recognition, 3D reconstruction, navigation, and obstacle avoidance, making indoor smart device smarter, safer, and accelerating the entry of smart devices into thousands of households.



AIGC

In the era of the explosive development of AIGC technology, a variety of generative large models have been born. Usually, these models are pre-trained based on a large amount of data, and SpatialVerse can provide training data resources including images, videos, models, and other elements.



Robotic Simulation

Robotic simulation can quickly, low-cost, and safely verify the product. SpatialVerse provides a simulation environment with high rendering and physical realism, and creates a simulation environment database by combining with specific simulation platforms to help enterprise robotic simulation.



Visualized Product Promotion

Visualized product promotion is a means by which customers can intuitively experience the allure of products. SpatialVerse harnesses its 3D data resource library and highly realistic environment presentation capabilities to empower demonstrations through data, thereby allowing users to experience product charm within a virtual environment.



XR

XR devices necessitate knowledge of real/virtual space objects and layouts. SpatialVerse enhances this via high-quality datasets, empowering researchers to craft precise spatial environments for improved XR user experiences. These datasets enable developers to create intelligent, responsive, and seamlessly blended immersive environments.



CLIENT SUCCESS STORY

Robotics Company Succeeds with SpatialVerse's Data Support

The Industry

- Intelligent Robot Research Institution

The Challenge

- Extensive Data Needs for Enhancing Product AI Capabilities
- Challenges include expensive data acquisition and lack of standardization

The Solution

- Data specialists catering to client needs
- Tailored data framework
- Testing and validation of data

The Results

- Data Expenses** Before utilizing SpatialVerse's data solutions, the client managed a large data team responsible for data collection, cleansing, and annotation. By adopting these solutions, the client experienced a remarkable reduction in overall data expenses, saving over 60% compared to previous costs.
- Data Diversity** Initially relying on real data, the client faced issues with data imbalance and difficulty in acquiring certain datasets. Leveraging SpatialVerse's extensive and ever-growing database, coupled with synthetic data advantages, the client was provided with complex data environments, producing large batches of unique datasets such as challenging lighting conditions and diverse surface textures.
- Data Quality** Establishing exclusive annotation and review frameworks according to client standards, each data resource was structured with descriptive systems including structural, positional, and status information. This facilitated large-scale model training, resulting in an increase in data annotation accuracy from 90% to 99%.
- Data Compliance** Faced with data sources originally scattered and products requiring global market coverage, the client encountered significant challenges in internal compliance management. SpatialVerse's database strictly adheres to legal regulations, offering complete authorization.

